Get Ready for Data Science in Practice

COMP 30780

Barry Smyth, Jan 2026

COMP 30780: Data Science in Practice

- This module is available to **3rd year Data Science students** in the School of Computer Science who are **not** taking up a place on the 5-month (15-credit) internship programme.

- All 3rd year Data Science students in the School of Computer Science are **auto-enrolled** into this module.

- If they secure an internship then they are un-enrolled from it at the end of February/early March.

Important Timings

- If you are hoping to pursue an internship, then contracts should be signed by the **first week in March (TBC)** with an approved start date at that time.

- Even if you have secured an internship, but your start date has not been approved by **the first week in March (TBC)** then you are obliged to participate fully in COMP30780 until your internship begins.

- It will not be possible to join COMP30780 late, because of a cancelled internship, which will leave you with a 15-credit gap.

Aim of this session

- Overview of the Module Structure & Objective
- Module Timetable & Key Events
- Deliverables & Grading
- Next Steps

Key People

- Barry Smyth (barry.smyth@ucd.ie)

- TA & Demonstrators
  - Manya Singh
  - Kripa Fathima Navas
  - Mossoun Franck Malick Jaures Ebiele
  - Reyhaneh Kheirbakhsh
  - Duc Anh Nguyen
  - Ifran Rahman Nijhum

Module Objectives

- To gain valuable **hands-on experience** developing a **group** (3-person) data science project from start to finish.
- Focus on **collaboration/reproducibility** best-practices for data science projects.
- Unique opportunity to develop and apply your data science skills and personal interests using a dataset(s) of **your choice/creation**.
- The module will introduce students new **computational**, **technical**, and **communication** challenges, combining **coding**, **writing**, and **presentation**.
- Emulating small-group data science collaboration and best practice.

Practical Skills

- **Identifying, preparing, pitching a suitable data science project.**
  - Identifying a suitable research question amenable to data-driven exploration.
  - Identifying/sourcing/assembling suitable dataset(s).
- **Data Wrangling/Analysis**
  - Preparing, cleaning, validating source data.
  - Using data to answer/explore appropriate research questions.
- **Data Presentation/Visualisation**
  - Summary statistics, grouping, correlation, causation.
  - Preparing analysis results and key conclusions.
- **Reproducibility Best Practice**
  - Environments, project structure, scripting, documentation, etc.

Module Structure

- Data Science Bootcamp
  - A sample data science project from start to finish.
  - Jupyter Notebook, Pandas, Numpy, Matplotlib etc.

} *Hands-on Intensive*

- Group-Based Data Science Project (Group x 3)
  - Identifying a suitable project topic.
  - Selecting/Creating & Preparing a Dataset(s)
  - Presenting your findings (report & seminar)

} *Personal Project Lab Support*

Core Deliverables

- **Annotated Dataset** – Generating/assembling a significant dataset; this may involve combining existing datasets in a novel way or creating new datasets from scratch.

- **Project Code** – Suitably structured/commented project code including data preparation, cleaning, analysis code in the form of a reproducible data processing pipeline.

- **Report** – Substantial report to motivate and explain the work, justifying the research question and presenting the core result/contribution.

- **Presentations** – Proposal/Final Project/Progress Updates (2 x week)

- **Participation** – Active participation in class/demonstrator meetings (attendance, discussion, etc.)

Workload Model (15 Credits)

- ***Full-Time***, Module requiring ~300 hours of effort across ~10 weeks (~***30hrs/wk***)

- ***Group-Based*** – all projects will usually involve a pair of students working together.

- ***Self-Defined*** – Each group is responsible for defining, planning, and managing their own project.

- ***Group work is challenging!*** Every effort will be made to understand the relative effort/contribution of different group members to ensure fair assessment.

Week Starting	Week Number	Actual Date	Day	Start Time	Length
02 Mar 2026	26	06 Mar 2026	Fri	16:00	50mins
23 Mar 2026	29	23 Mar 2026	Mon	10:00	170mins
	29	24 Mar 2026	Tue	10:00	170mins
	29	25 Mar 2026	Wed	10:00	170mins
	29	26 Mar 2026	Thu	10:00	170mins
	29	27 Mar 2026	Fri	10:00	170mins
30 Mar 2026	30	30 Mar 2026	Mon	10:00	110mins
	30	02 Apr 2026	Thu	10:00	110mins
06 Apr 2026	31	10 Apr 2026	Fri	10:00	110mins
13 Apr 2026	32	13 Apr 2026	Mon	10:00	110mins
	32	17 Apr 2026	Fri	10:00	110mins
20 Apr 2026	33	20 Apr 2026	Mon	10:00	110mins
	33	24 Apr 2026	Fri	10:00	110mins
27 Apr 2026	34	27 Apr 2026	Mon	10:00	110mins
	34	01 May 2026	Fri	10:00	110mins
04 May 2026	35	08 May 2026	Fri	10:00	110mins

Group Assignment Session & Preparation Instructions

In-Person Bootcamp (Mon-Friday, 10 am - 1pm)

4 weeks of project planning & dataset collection

Proposal Presentation

RQ1 / Status Update

RQ2 / Status Updates

RQ3 / Status Updates

4 weeks of analysis.

Final Technical Week

Final Presentation (Report Due on May 15

2 weeks for presentation & writing.

The First 4 Weeks

- **Project Proposal (March 6 - April 2; note:** *includes Field Study weeks***)**
  - Get to know your group! Practice working as a team. Agree a schedule for meetings etc.
  - Explore/develop preliminary ideas with your group partners (March 9-20)
  - Identify lead/backup project idea. Assemble dataset. Discuss during bootcamp week (March 23-27)
  - Finalise project proposal (research questions, dataset gathering, class presentation) (March 30 - April 2)
  - *Probably the most important phase of the project and the one that most correlates with success!*

- **Bootcamp Week (March 23-27) - In-Person, all week!**
  - Intensive in-class sessions, typically from 10 am-1 pm each day.
  - Focus on a demonstration data science project developed in-class.
  - Support group project planning; in-class discussion of project ideas etc.
  - Check-in on group project progress.

A Typical Project Week (April 8 – …)

- **Regular Project Weeks**
  - Mondays @10-12, In-person project planning stand-ups .
  - Wednesday Demonstrator Meetings
  - Fridays @10-12, In-person project progress stand-ups.

- **Presentation & Deadline Weeks**
  - Project Proposal Presentation (~10 mins + Q/A) (April 2)
  - Final Project Presentation (~15 mins + Q/A) (May 8)
  - Final Report (25-40 pages, Overleaf/Latex) (May 15)

Assessment

- **100% continuous assessment with a final grade based on:**
  - Conduct & contribution (10%) **in-class / demonstrator meetings etc.*
  - Project Proposal (20%) **Based on your final proposal presentation.*
  - Project code (25%)
  - Final report (25%)
  - Final presentation (20%)

- **Group contributions**
  - Each group member is expected to make an equal and appropriate contribution and efforts will be made to evaluate individual contributions.
  - Early interventions are key when issues/conflicts do arise; please act early by escalating if you cannot address between yourselves.

Next Steps

- Make sure to attend the **Group Assignment** session session on **March 6 at 4 pm**. Groups will be assigned at this session and further details will be available on Brightspace in the coming weeks.

- Do **try to find a group partner** beforehand and start brainstorming project ideas. We will be in touch before March 6th to collect preferred group partnerships.

- If you intend to take up an internship then please work to ensure you secure it in time and **no later than first week in March**; if you have any doubts then make sure you are prepared for COMP30780 as your backup plan.

- Lots of material uploaded to Brightspace. Check out slides/videos/past bootcamps etc. in My Learning section. **Required reading/viewing before Match 6!**